

The declining Atlantic overturning freshwater transport at 34.5°S is not explained by a declining overturning transport

Abstract

Freshwater transport by the overturning at the southern boundary of the Atlantic, F_{ov}^S , is the standard indicator of whether the salt-advection feedback is self-reinforcing, and its recent decline in ocean reanalyses is increasingly read as a warning that the overturning is weakening. We show that it is not. Once the net throughflow is removed, F_{ov}^S factorises exactly as $-T \Delta S / S_0$, with T the northward-limb volume transport and ΔS the transport-weighted salinity difference between the limbs, both measured directly from the section. At 34.5°S F_{ov}^S changes at 36% (ORAS5) and 52% (GLORYS12V1) of its own magnitude per decade while T changes at 2.6% and 0.6%. The transport contributes 1% to 18% of the trend depending on how the overturning is defined, in most configurations with the opposite sign, and stays below 25% in 97% of all admissible epoch pairs in the two declining products. What the decline records instead is a widening limb salinity contrast, carried by the northward limb; the EN4 observational analysis reproduces that asymmetry in a salinity field no model integrates. The products disagree on the rest: ECCO-V4r4 has no trend to attribute, and in ORAS5 the split between salinity and flow structure is not resolved, with a third of the widening attributable to the flow.

Introduction

The Atlantic meridional overturning circulation transports salt as well as heat, and the resulting salt-advection feedback is the classical route to multiple equilibria in this basin^{1–3}. The diagnostic used to test whether that feedback is self-reinforcing is F_{ov}^S , the freshwater transport carried by the overturning across the southern boundary^{4–6}. When $F_{\text{ov}}^S < 0$ the overturning imports freshwater into the basin, so a weakening circulation freshens the Atlantic and weakens it further; when $F_{\text{ov}}^S > 0$ the same perturbation is damped. In box models and in coarse-resolution ocean models the sign of F_{ov}^S separates a monostable from a bistable regime, and its value has been used to argue that most climate models are biased towards the stable side^{7–9}.

That correspondence is not exact. Extended box models¹⁰ and hysteresis experiments in a state-of-the-art coupled model¹¹ show that the collapsed branch can persist well outside the $F_{\text{ov}}^S < 0$ range, so the sign indicates where the freshwater budget sits rather than proving that a second equilibrium exists. The *transient* behaviour of F_{ov}^S is a further step removed. F_{ov}^S reflects the salinity contrast between the limbs at the section, which responds to North Atlantic anomalies only after they have been advected south, and an AMOC collapse in a coupled model need not coincide with a minimum in F_{ov}^S ^{12,13}. Despite this, a declining F_{ov}^S is increasingly read as a physics-based early warning that the circulation is weakening^{14–16}, and it is that inference we test.

The inference is not safe, for a reason visible in the definition. Write F_{ov}^S as the product of an overturning transport and a salinity contrast. Wherever $F_{\text{ov}}^S < 0$ the contrast is positive, so a weakening overturning *raises* F_{ov}^S rather than lowering it, and a falling F_{ov}^S cannot by itself be evidence of a weakening circulation¹³. (The argument applies only where $F_{\text{ov}}^S < 0$ throughout; ORAS5 has 7 years of 68 in which it does not hold.) Whether the observed decline is carried by the transport or by the salinity field is an empirical question, and it determines how much of the recent literature on F_{ov}^S trends can be interpreted at all.

Answering it requires a factorisation whose two factors are separately measurable. Defining the salinity contrast as $-S_0 F_{\text{ov}}^S / \Psi$ for some scalar overturning strength Ψ does not qualify: the second factor is then the first divided out, it absorbs every change in the vertical structure of the velocity as well as every change in salinity, and the resulting attribution is fixed by algebra rather than by data. We therefore use the limb factorisation. Removing the net section throughflow, which any transport-consistent estimate of F_{ov}^S requires, forces the depth integral of the section velocity to vanish, so the northward and southward limbs carry equal and opposite volume transport T , and splitting the depth integral at the sign changes of the velocity gives

$$F_{\text{ov}}^S = -\frac{1}{S_0} T \Delta S, \quad \Delta S = S_{\uparrow} - S_{\downarrow}, \quad (1)$$

exactly and with no residual, where $S_{\uparrow}$ and $S_{\downarrow}$ are the transport-weighted mean salinities of the northward and southward limbs. Equation (1) is not a linearisation and not a fit. Both factors are computed directly from the section profile, T is linear in the velocity field and so carries none of the sampling bias of a streamfunction maximum, and ΔS is a salinity difference that can be plotted, compared between products, and in principle measured by hydrography.

We evaluate this at 34.5°S in three ocean reanalyses recomputed here from monthly fields, ORAS5, GLORYS12V1 and ECCO-V4r4, spanning two ocean models and three assimilation methods, and we ask what the answer implies for the interpretation of CMIP6 freshwater transport biases.

Results

A negative freshwater transport, and the two factors that set it

Recomputed from monthly fields on the contiguous Atlantic section (Methods), the record-mean F_{ov}^S at 34.5°S is negative in all three reanalyses (Fig. 1a,b): -0.038 Sv in ORAS5 over 1958–2025, -0.084 Sv in GLORYS12V1 over 1993–2025 and -0.132 Sv in ECCO-V4r4 over 1992–2017. The sign is not marginal. It holds in 61 of 68 annual means in ORAS5, 32 of 33 in GLORYS12V1 and 26 of 26 in ECCO, and the 95% circular block-bootstrap interval excludes zero in every product at every block length from 2 to 20 years, so the result does not depend on that choice. Because a mean over a trending record is not an estimate of any well-defined population quantity, we also give the most recent decade, which is the quantity a reader interested in the present state should use: -0.088 [-0.094 , -0.080] Sv in ORAS5 over 2016–2025, -0.132 [-0.141 , -0.121] in GLORYS12V1 and -0.128 [-0.136 , -0.119] in ECCO over 2008–2017.

A fourth product, SODA 3.15.2, gives -0.009 Sv with an interval that includes zero, but we do not count it either way. The archive we can obtain provides annual rather than monthly fields, and F_{ov}^S is bilinear in velocity and salinity, so forming it from annual means discards the within-year covariance. Measured directly in ECCO, where both routes are available, that omission shifts F_{ov}^S by $+0.012$ Sv, larger in magnitude than SODA’s entire record mean. The same test gives $+0.000$ Sv in ORAS5 and -0.001 Sv in GLORYS12V1, so the bias is product-dependent and small in the two eddy-permitting products. SODA can neither support nor refute the sign, and we say so rather than counting it.

The magnitudes span a factor of three across the three products, which is larger than any within-product interval. The reanalyses differ more about how much freshwater the overturning carries than about which way it carries it, and only the second statement is robust.

Equation (1) splits each of these numbers into a transport and a salinity contrast. The northward limb carries 18.7 Sv in ORAS5, 21.3 Sv in GLORYS12V1 and 14.6 Sv in ECCO, against limb salinity contrasts of $+0.068$, $+0.136$ and $+0.307$ PSU. (The factorisation is exact year by year, so the product of the record means differs slightly from the mean of the products, by 6% in ORAS5.) Almost the entire three-product spread in F_{ov}^S is in ΔS , which varies by a factor of 4.5, not in T , which varies by a factor of 1.5. Whatever the reanalyses constrain at this section, it is the transport far better than the salinity structure.

The decline outruns the transport by an order of magnitude

ORAS5 and GLORYS12V1 both decline over their own records, by -1.37 mSv per year ($p < 10^{-4}$, $N_{\text{eff}} = 23.7$) and -4.36 mSv per year ($p = 0.017$, $N_{\text{eff}} = 7.2$). ECCO does not decline at all ($+0.11$ mSv per year, $p = 0.76$, $N_{\text{eff}} = 23.1$, minimum detectable trend 0.75 mSv per year). Over the 1993–2017 window all three share, no product has a significant F_{ov}^S trend, but that is a statement about power rather than about the ocean: the smallest trend those 25 years could have resolved is 1.3 mSv per year in ORAS5 and 7.1 in GLORYS12V1, and the GLORYS12V1 point estimate over the common window (-4.76) is larger in magnitude than over its own record. The northward-limb transport behaves differently over that same window, and in the opposite direction to the paper’s framing: it declines in all three products, by -0.97 Sv per decade in ORAS5 ($p = 0.052$, $N_{\text{eff}} = 10.1$), -0.75 in GLORYS12V1 ($p = 0.036$, $N_{\text{eff}} = 14.3$) and -0.35 in ECCO ($p = 0.062$, $N_{\text{eff}} = 21.3$). Over the one window in which the three products are directly comparable it is the transport, not F_{ov}^S , that has the near-significant trends.

The cleanest statement of the result needs no ratio at all. Expressed as fractional rates, F_{ov}^S is changing at -36% of its own magnitude per decade in ORAS5 and -52% in GLORYS12V1, while the transport that carries it changes at -2.6% and -0.6% per decade. Over the Argo era the same comparison gives -35% against -0.7% and -20% against $+0.8\%$. In those windows the two quantities are changing at rates that differ by between one and two orders of magnitude, which is the argument in its simplest form and is free of any denominator that might approach zero. The separation is not uniform across every window: over the common 1993–2017 window, where the F_{ov}^S trend is not resolvable and the transport trend nearly is, the ORAS5 ratio narrows to 2.5. Across

the four windows we examine, the transport rate is between 1% and 132% of the F_{ov}^S rate in the two declining products. The wide end comes entirely from windows in which the F_{ov}^S trend is not resolvable: 40% from the common window in ORAS5 and 132% from the eleven-year clean window in GLORYS12V1, where the F_{ov}^S trend has the wrong sign and $p = 0.79$. Over the windows in which the denominator is resolvable the range is 1% to 7%.

Splitting the trend by equation (1) puts numbers on it (Fig. 2b). In ORAS5 the total is -1.37 mSv per year, of which the transport term contributes $+0.09$ [$+0.06$, $+0.13$], the salinity term -1.46 [-1.77 , -1.14] and a covariance residual $+0.002$, 3% of the transport term. In GLORYS12V1 the total is -4.36 , the transport term $+0.05$ [-0.12 , $+0.21$], the salinity term -4.36 [-5.78 , -2.94] and the residual -0.05 , which is 108% of the transport term and larger than it. The transport share of the trend is 0.068 [0.046 , 0.094] in ORAS5 and 0.012 [0.001 , 0.051] in GLORYS12V1; assigning the residual entirely to the transport instead of dropping it moves the GLORYS12V1 share to 0.001 and flips its sign, so for that product the honest statement is that the transport contribution is not distinguishable from zero rather than that it is positive. The transport term opposes the total with probability 1.00 in ORAS5 over its own record but only 0.73 in GLORYS12V1, and 0.36 in GLORYS12V1 over the Argo era, where its point estimate (-0.09 mSv per year) has the same sign as the total. The magnitude claim is robust; the sign claim is not universal and we do not make it universal.

What the salinity term represents is unusually specific (Fig. 3a). The contrast ΔS widens because the northward limb becomes saltier while the southward limb does not change. $S_{\uparrow}$ rises by $+0.025$ PSU per decade in ORAS5 ($p < 10^{-4}$, $N_{\text{eff}} = 24.2$) and $+0.071$ in GLORYS12V1 ($p = 0.017$, $N_{\text{eff}} = 6.9$, which is under five degrees of freedom and should be read accordingly), whereas $S_{\downarrow}$ changes by -0.002 ($p = 0.06$) and -0.001 ($p = 0.24$) over the same records. The southward limb is not uniformly flat: over the matched 1993–2017 window the ORAS5 value is -0.003 and does reach significance ($p = 0.006$). The asymmetry rests on magnitude, $S_{\uparrow}$ moving six times faster, rather than on $S_{\downarrow}$ being exactly stationary. Over the Argo era the two products converge, to $+0.051$ ($p = 0.009$) and $+0.035$ ($p = 0.005$); note that this is a rise for ORAS5 and a halving for GLORYS12V1, so the agreement improves while the GLORYS12V1 rate falls. Over the matched 1993–2017 window neither is individually significant ($+0.019$, $p = 0.11$, $N_{\text{eff}} = 15.8$; $+0.083$, $p = 0.10$, $N_{\text{eff}} = 5.2$) and they differ by a factor of four.

The depth-resolved salinity trends behind these numbers disagree far more than the limb-weighted quantities do (Fig. 3b). Through the upper 500 m GLORYS12V1 trends at $+0.10$ to $+0.13$ PSU per decade against a layer-mean $+0.009$ in ORAS5, which crosses zero near 200 m, so the two differ by more than an order of magnitude and are jointly significant over only a narrow band near 600 m. That the transport-weighted limb salinities agree to within a factor of 2.3 over 1993–2025 while the raw profiles disagree by more than ten is an argument for weighting by the transport that actually carries the salt rather than by depth.

ECCO is the counterexample, and against GLORYS12V1 it is a real one rather than a lack of power. Its limb salinities are flat ($+0.005$ PSU per decade, $p = 0.55$, $N_{\text{eff}} = 14.9$). On the matched 1993–2017 window its minimum detectable $S_{\uparrow}$ trend is 0.020 PSU per decade, well below the GLORYS12V1

rate of +0.083 over the same years, so ECCO could have resolved that salinification and did not. Against ORAS5 the comparison is inconclusive: the ORAS5 common-window rate, +0.019 PSU per decade, is itself below ECCO's detection limit.

Is the widening contrast a water mass, or a redistribution?

ΔS is transport-weighted, so it changes if the salinity field changes, and also if transport is redistributed in depth at a completely fixed salinity field. The two have to be separated before the change can be called a water-mass signal. Recomputing ΔS with the salinity profile frozen at its record mean, and again with the velocity profile frozen, does that at the full vertical resolution of the section, and the answer differs between products.

In GLORYS12V1 the change is almost all salinity: the salinity-only trend is +0.068 PSU per decade ($p = 0.048$, $N_{\text{eff}} = 4.8$) against a full trend of +0.072, a share of 0.94, and the velocity-only trend is +0.007, not significant and close to its own detection limit ($p = 0.22$, $N_{\text{eff}} = 22.0$, detection limit 0.011). In ORAS5 the partition is not resolved. The point estimates give a salinity share of 0.66 and a velocity share of 0.39, so roughly a third of the widening is a change in the vertical structure of the flow rather than in the water masses. But the salinity-only series is so persistent (lag-1 autocorrelation 0.93) that its effective sample size falls to the floor of 3 imposed by the significance test, and the smallest trend it could have resolved is 0.199 PSU per decade, eleven times the estimate itself. Its $p = 0.45$ therefore carries no information. The velocity-only term is the resolvable one (+0.011 PSU per decade, $p = 0.0002$, $N_{\text{eff}} = 31.9$, detection limit 0.005), so what can be said of ORAS5 is that a flow-structure contribution is definitely present, not that the salinity contribution is absent. In GLORYS12V1 the salinity-only estimate also sits close to its own detection limit (+0.068 against 0.066, $N_{\text{eff}} = 4.8$). In ECCO neither term is resolvable (−0.006, detection limit 0.010; +0.009, detection limit 0.027) and the full trend is indistinguishable from zero. The two shares sum to slightly more than one because the split leaves a small covariance residual, −5% in ORAS5 and −4% in GLORYS12V1.

The coarse component of that redistribution can be isolated further. The limbs are defined by the sign of the section velocity, and that sign changes more than once: besides the upper cell there is a northward abyssal limb of Antarctic Bottom Water carrying 3.8 Sv in ORAS5, 4.8 in GLORYS12V1 and 2.2 in ECCO, that is 20%, 23% and 15% of the northward limb, and fresher than the upper cell by 0.21 to 0.49 PSU (Extended Data Fig. 2a,c). Writing $S_{\uparrow} = \sum_i w_i S_i$ over the sub-limbs and holding either the weights or the sub-limb salinities at their record means splits it. The partition term is small and of the opposite sign to the total in both declining products: in ORAS5 the change in $S_{\uparrow}$ of +0.027 PSU per decade is a water-mass term of +0.030 and a partition term of −0.003, a share of 0.12; in GLORYS12V1, +0.072 splits into +0.073 and −0.002, a share of 0.03. So the upper-to-abyssal redistribution is not what drives the ORAS5 velocity term; that lives within the upper limb.

ECCO is the one product in which the coarse redistribution is significant and active. Its upper-cell transport falls at −0.60 Sv per decade ($p = 0.011$) while its abyssal limb rises at +0.36 ($p = 0.041$),

and its upper-cell salinity rises at +0.022 PSU per decade ($p = 0.025$), the same sign as the other two products. That last p is basis-dependent: identifying the sub-limbs on annual-mean profiles instead gives +0.018 and $p = 0.18$, so the ECCO cancellation is suggestive rather than established. ECCO’s flat whole-limb salinity is therefore the sum of a positive water-mass term and a redistribution that cancels it, not an absence of change, and the confound this section exists to test is demonstrably real even though it is not what dominates elsewhere.

Excluding the abyssal cell altogether points the same way as the sub-limb split. Truncating the section at the deep zero crossing and re-applying the throughflow removal within the truncated layer, so the factorisation is again exact, gives upper-cell $S_{\uparrow}$ trends of +0.033 PSU per decade in ORAS5 ($p < 10^{-4}$) and +0.086 in GLORYS12V1 ($p = 0.033$), larger than the whole-limb values, with transport shares of 0.112 and 0.019.

An observation-only test

The claim that the northward limb is becoming saltier can be tested in a salinity field that no ocean model integrates forward. We take EN4.2.2, an objective analysis of profile observations, hold the velocity structure frozen at each reanalysis record-mean profile, and let only the observed salinity vary over 2005–2024 (Methods). Any trend is then a trend in observed water masses under a fixed circulation. EN4 is on whole-degree latitudes, and we use the 34°S row: the 35°S row lies south of Cape Agulhas, where Africa no longer closes the basin and a contiguous section runs into the Indian Ocean.

The limb asymmetry is reproduced, and more clearly than in the reanalyses’ own statistics. $S_{\uparrow}$ rises at +0.026 PSU per decade ($p = 0.0005$) under the ORAS5 velocity structure and +0.028 ($p = 0.0003$) under the GLORYS12V1 structure, while $S_{\downarrow}$ is flat in both (+0.002, $p = 0.09$), and the contrast widens at +0.024 to +0.026 PSU per decade ($p = 0.0001$). The sign, the limb asymmetry and the significance therefore survive in a salinity field that no assimilating model has touched.

The observed salinity field also supplies an observational estimate of F_{ov}^S itself. Combined with the frozen transport it gives −0.080 Sv under the ORAS5 velocity structure and −0.128 Sv under the GLORYS12V1 structure, both firmly negative. This is not an independent measurement, since the transport is taken from the reanalyses, but it shows the negative sign does not depend on the reanalysis salinity fields. It also sharpens the disagreement about magnitude: under ORAS5’s own velocity structure the observed salinity field returns −0.080 Sv against ORAS5’s own −0.038 Sv, because EN4’s limb contrast at this section is 0.160 PSU against ORAS5’s 0.068.

The rate is a different matter. Over the identical 2005–2024 window the reanalyses give $S_{\uparrow}$ trends of +0.056 (ORAS5) and +0.035 (GLORYS12V1) PSU per decade, against +0.026 and +0.028 from the observations. ORAS5 runs about twice the observation-only rate while GLORYS12V1 is within 30% of it. Part of that gap is expected by construction, since EN4 relaxes towards climatology where profiles are sparse and the core Argo array does not sample below 2000 m, so the observation-only estimate is a lower bound. The direction is robust; the ORAS5 rate should not be taken at face value.

The attribution does not depend on the choices we made

Epoch selection. Rather than quoting hand-picked epochs, we evaluate the symmetric decomposition of equation (1) for every non-overlapping early and late epoch pair each record supports, at epoch lengths of 10, 13 and 15 years: 2901 usable pairs in ORAS5 and 151 in GLORYS12V1. These are not independent samples, and we treat them as a sensitivity analysis over window choice rather than as evidence from a large sample. The median transport share is 0.075 in ORAS5 and 0.038 in GLORYS12V1, the 95th percentile is 0.219 and 0.088, the maximum is 0.90 and 0.12, and the share is below 0.25 in 96.8% and 100% of pairs (Fig. 2c). The transport term opposes the total in 96.8% and 94.7% of pairs. The symmetric form uses the mean of the two epochs as its coefficient, so it has no cross term and does not depend on which epoch is the base point; the early-base and late-base conventions bracket it, with median shares of 0.012 and 0.141 in ORAS5.

The definition of overturning strength. Substituting the depth-space streamfunction maximum for the limb transport, either integrated from the surface or with the integration restarted at 250 m to exclude the wind-driven surface cell, changes the mean overturning strength substantially, from 18.7 to 14.6 to 8.6 Sv in ORAS5, and raises the share to 0.111 [0.076, 0.151] and 0.183 [0.137, 0.240]; in GLORYS12V1 it becomes 0.014 and 0.040. The share therefore rises monotonically as the definition of the overturning narrows, by a factor of 2.4 across the three definitions computed on this common basis (0.076, 0.111 and 0.183 for ORAS5), and the full range across every test in which the share is identified, excluding the census tail, is 1% to 18% with individual intervals reaching 27%; the two largest values with a resolvable denominator are 0.18 over 1993–2025 in ORAS5 and 0.183 for the 250 m streamfunction definition over its full record. It remains a minority contribution throughout, but it is not definition-independent, and we note that the factorisation we advocate gives a smaller share than the construction it replaces.

The observing system. The salinity field at 34.5°S was barely constrained before Argo, so we repeat everything from 2004 onwards. The GLORYS12V1 decline survives (-2.28 mSv per year, $p = 0.038$) and its transport share rises, to 0.038 [0.004, 0.268], while remaining a minority. GLORYS12V1 has no documented stream join, but its record crosses the Argo transition, and applying the same level-shift model at 2004 shows that most of its signal is a step rather than a trend: the residual trend falls to -1.90 ± 1.07 mSv per year ($p = 0.098$, not significant) against a step of -0.061 ± 0.022 Sv ($p = 0.014$), and the limb contrast behaves the same way ($+0.029 \pm 0.015$ PSU per decade, $p = 0.071$, against a step of $+0.105 \pm 0.031$, $p = 0.004$). The attribution is unaffected, the step-adjusted transport share being 0.045 and the step-adjusted fractional rate 23% per decade rather than 52%, but the GLORYS12V1 *decline* is a level shift at the arrival of Argo and should not be read as a trend.

ORAS5 is also a splice, the consolidated reanalysis to 2014 and the operational stream thereafter, and here too the test is weaker than it looks. Fitting a trend together with a level shift at 2015 gives, over the full record, a step in F_{ov}^S of -0.017 ± 0.011 Sv ($p = 0.15$); the corresponding step in the limb salinity is larger relative to its own signal and is significant, $+0.038 \pm 0.017$ PSU ($p = 0.029$), and allowing for it reduces the $S_{\uparrow}$ trend from $+0.025$ to $+0.020$ PSU per decade. Refitting over 2004–2025 alone gives a trend of -1.36 ± 1.53 mSv per year that is not significant, against -2.46 when the step is ignored. The ORAS5 Argo-era decline does not survive fitting the splice. Over the full

record the step-adjusted trend is -1.17 ± 0.21 mSv per year ($p < 10^{-4}$), essentially identical to the consolidated-stream-only value of -1.17 ($p = 5 \times 10^{-5}$), whose transport share is 0.050 [0.024, 0.083]. So roughly 15% of the raw ORAS5 full-record decline is the splice, the decline survives correcting for it over the full record, and the attribution is unaffected; but the 22-year Argo window alone cannot separate the two.

The one window with neither confound. 2004–2014 is inside the Argo era and inside the ORAS5 consolidated stream at once. It is short, and it settles nothing: over it the ORAS5 fractional-rate ratio is 0.152 with an F_{ov}^S trend that is not resolvable, and GLORYS12V1 gives an F_{ov}^S trend of the opposite sign (+7% per decade, $p = 0.79$) against a transport falling at -9% per decade ($p = 0.0005$), so its fractional-rate ratio is 1.32; ECCO gives 0.30. Eleven years cannot separate these quantities, and we report the window rather than omit it because it is the only one that satisfies both homogeneity conditions.

The removal of the net transport. Taking the net section transport out in proportion to $|V|$, so that it is removed where the flow is rather than uniformly over the area, changes T by less than 0.6% and leaves the attribution intact: the transport share becomes 0.078 in ORAS5 and 0.012 in GLORYS12V1, against 0.068 and 0.012 for the scheme used throughout.

The window. Over 1993–2025, the longest window the two eddy-permitting products share, both limb-salinity trends are resolvable ($+0.031$ PSU per decade, $p = 0.004$, in ORAS5 and $+0.071$, $p = 0.017$, in GLORYS12V1, a factor of 2.3 apart rather than the factor of four over the shorter three-product window). The ORAS5 transport share on that window is 0.180, and still a minority.

The section. Recomputing the whole diagnostic at 25, 28, 30, 32 and 34.5°S over the matched 1993–2025 window (Fig. 1c), the record mean is negative at every latitude in both products, and its interval excludes zero everywhere except ORAS5 at 25°S (-0.004 Sv). The magnitude varies by a factor of 22 in ORAS5 across the full band and by 1.5 in GLORYS12V1, and at 25°S the two products disagree by a factor of 22, a larger disagreement than any we report at 34.5°S . (The two factors, 22.5 and 22.6, are close by coincidence.) Excluding 25°S , the ORAS5 range narrows to a factor of two. The negative sign is established in ORAS5 only poleward of about 28°S . We use 34.5°S because that is where the observing array sits and where the diagnostic is defined.

Two further quantities bound what the diagnostic means. The net volume transport across the section is -1.2 Sv in ORAS5, -1.2 in GLORYS12V1 and -1.0 in ECCO, and removing it in order to define an overturning component shifts F_{ov}^S by $+0.009$, $+0.008$ and $+0.007$ Sv. Separately, the overturning is not the only freshwater carrier at this section: the azonal component is $+0.306$ Sv in ORAS5, $+0.304$ in GLORYS12V1 and $+0.155$ in ECCO, opposite in sign to F_{ov}^S and, in the eddy-permitting products, several times larger. Its trend is also opposite and, over each product’s own record, significant: $+1.84$ mSv per year in ORAS5 ($p = 0.0002$, $N_{\text{eff}} = 21.8$) and $+6.68$ in GLORYS12V1 ($p = 0.049$, $N_{\text{eff}} = 8.5$, so marginal on few effective degrees of freedom), both larger in magnitude than the F_{ov}^S trends they accompany, against $+0.19$ ($p = 0.95$) in ECCO, where the detection limit is 14 mSv per year and the null is uninformative. The total freshwater transport across 34.5°S is therefore not moving the way F_{ov}^S alone implies, and a reader assessing whether the basin is moving deeper into the self-reinforcing regime must weigh both. F_{ov}^S is a stability diagnostic

defined on the overturning component; it is not the freshwater budget of the basin.

Implications for climate models

CMIP6 historical simulations sit on the other side of zero, but not by a majority that can be tested. Over 1950–1980, 16 of 25 models have a positive mean F_{ov}^S at 34.5°S, with a median of +0.053 Sv against a range of −0.205 to +0.405 Sv (Fig. 4a). That count is not distinguishable from a coin flip (binomial $p = 0.23$), and taking one member per model family leaves 18 independent models of which 10 are positive ($p = 0.81$), or 9 if the other member of the one internally inconsistent family is chosen ($p = 1.00$). The defensible statement is about location rather than majority: the model median is +0.053 Sv while all three recomputed observational estimates lie between −0.038 and −0.132 Sv, below 72%, 80% and 88% of the models respectively. This confirms a bias documented previously^{7–9} and benchmarks it against three consistently computed estimates; it does not discover it. The model means are over 1950–1980 while the reanalyses cover modern decades, and since ORAS5 declines by 1.37 mSv per year the two windows are not equivalent; the comparison should be read as a statement about a persistent model offset, not as a like-for-like difference of contemporaneous states.

Whether that bias matters for projected rates is a separate question, and the test that has been used to answer it has no power. Correlating model mean F_{ov}^S against a 19-year overturning trend at 26.5°N over 2005–2023 gives $r = -0.04$ ($p = 0.88$) across 16 models, but the smallest correlation that sample could reach significance with is $|r| = 0.50$, and 19-year overturning trends within a single model’s large ensemble already span −3.5 to +2.4 Sv per decade across 50 members of MPI-ESM1-2-LR. A null from such a test is guaranteed by internal variability whatever the forced relationship. The counting statistic behaves the same way: 2 of 6 models with negative F_{ov}^S weaken faster than GLORYS12V1 (whose own rate at 26.5°N is −0.70 Sv per decade) against 6 of 10 with positive F_{ov}^S , a Fisher exact p of 0.61, with the same counts against the RAPID rate.

Repeating the test on the forced response, where the signal is large, remains negative but is now informative. Across 21 models with historical plus ssp585 projections to 2100, mean F_{ov}^S and the overturning change from 1850–1900 to 2081–2100 give $r = -0.20$ ($p = 0.40$, 95% CI −0.58 to +0.26) against a smallest reachable $|r|$ of 0.43 (Fig. 4b); on fractional rather than absolute change $r = +0.30$ ($p = 0.18$). Leaving out one model at a time moves the first between −0.41 and −0.05 and the second between +0.06 and +0.45, so neither is stable.

The reason the two have opposite signs is itself the one significant result in this comparison. Model mean F_{ov}^S and the pre-industrial overturning strength are strongly correlated ($r = +0.73$, $p = 1.5 \times 10^{-4}$, 95% CI +0.44 to +0.89, $n = 21$): models with a stronger overturning carry a more positive F_{ov}^S , which is what equation (1) predicts given that these models mostly carry $\Delta S < 0$. Absolute overturning change is mechanically tied to the base state while fractional change is not, which is why the two rate tests disagree in sign. It also suggests that the CMIP6 F_{ov}^S bias is in part a symptom of the overturning strength bias rather than an independent error in the freshwater budget.

The observed rate is itself weakly constrained. Over 2005–2023 the RAPID array gives −0.89 Sv per decade at 26.5°N, with an unadjusted p of 0.093 that becomes 0.21 once the effective sample size is

340 accounted for ($N_{\text{eff}} = 11.9$). The direct observations do not yet establish a significant decline over
341 the era in which the reanalyses are best constrained^{17,18}.

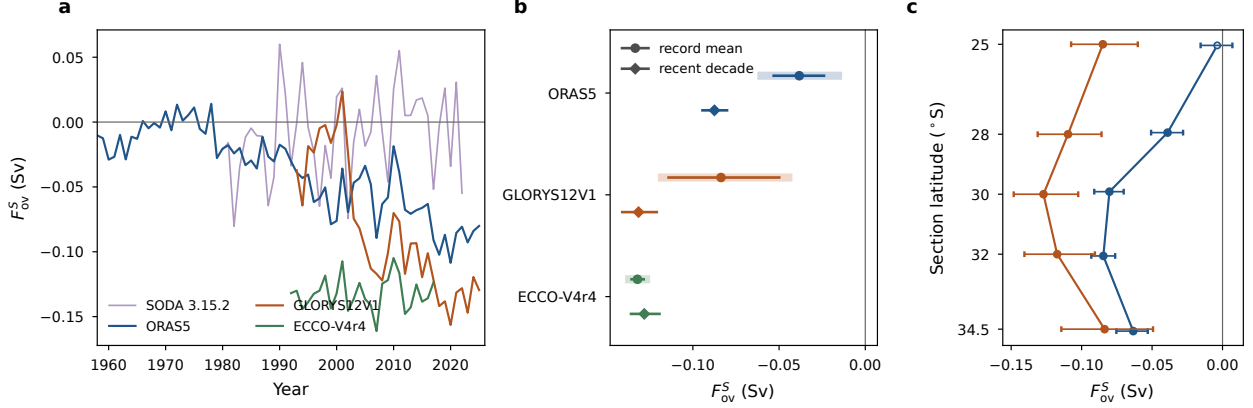


Figure 1. The sign of the overturning freshwater transport at 34.5°S. **a**, Annual-mean F_{ov}^S . ORAS5, GLORYS12V1 and ECCO-V4r4 are computed here from monthly fields on the contiguous Atlantic section; SODA 3.15.2 is carried over from an annual-field pipeline, is not on the same basis, and is not used in any quantitative claim. **b**, Record mean (circles) and most recent decade (diamonds) with 95% circular block-bootstrap intervals, at a 5-year block for the record mean and a 3-year block for the decade. The pale bar spans the intervals obtained for block lengths of 2 to 20 years. Between-product spread is 0.094 Sv, wider than the widest of these intervals (0.073 Sv, GLORYS12V1 across block lengths) and 1.5 times the 5-year intervals quoted in the text. **c**, Mean F_{ov}^S over 1993–2025 against section latitude, with the same intervals. The open marker is the one case whose interval includes zero.

Discussion

Two statements about F_{ov}^S have been run together, and they need to be separated. The first is that the Atlantic freshwater budget at 34.5°S sits on the side where the salt-advection feedback is self-reinforcing. Three reanalyses spanning two ocean models and three assimilation methods agree on that, with intervals clear of zero at every block length we tried, at every latitude from 28 to 34.5°S in the two products run on the latitude ladder, and in the most recent decade of each record. The second is that the observed decline in F_{ov}^S measures a weakening overturning. That statement does not survive contact with the same data.

The reason is a rate comparison rather than a ratio. In the two products that decline, F_{ov}^S is changing at 36% and 52% of its own magnitude per decade while the volume transport that carries it changes at 2.6% and 0.6%. Passed through equation (1) this makes the transport contribution between 1% and 18% of the trend depending on how the overturning is defined and over which window, with individual intervals reaching 27%, and in most configurations it has the opposite sign to the total. It is a minority contribution under every test we ran in which the share is identified, though it is not definition-independent: it rises systematically as the definition of the overturning narrows from the whole northward limb to the streamfunction maximum. Studies that read a falling F_{ov}^S as a proxy for AMOC weakening, including the early-warning arguments built on the F_{ov}^S minimum¹⁴ and an earlier, unpublished version of our own analysis of these data, have attributed to the circulation a change that the circulation did not make.

What did make it is a widening of the salinity contrast between the limbs, carried by the northward

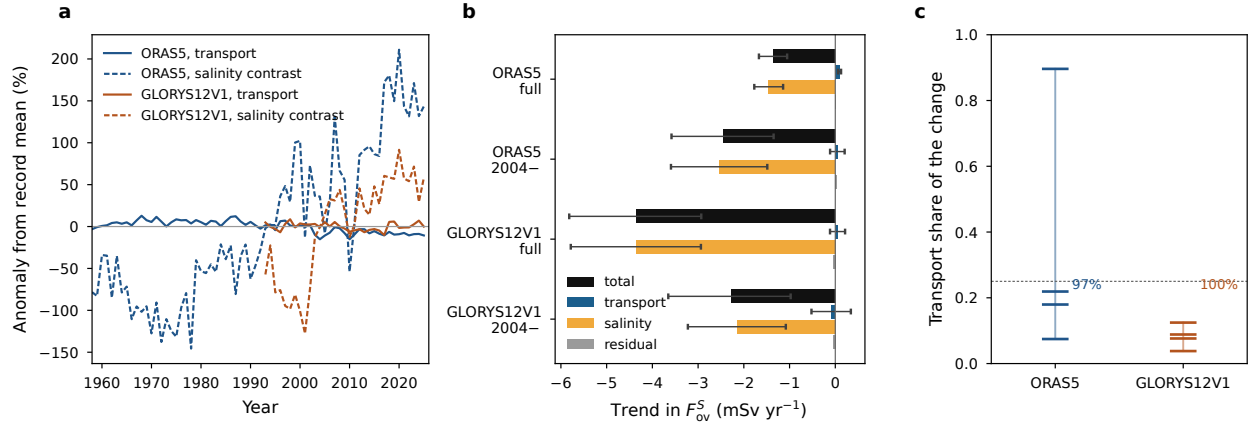


Figure 2. The decline is carried by the salinity contrast, not by the transport. **a**, The two factors of equation (1), each as a percentage anomaly from its own record mean, so that a factor which barely moves can be compared with one that moves a great deal. The ORAS5 contrast passes close to zero early in the record, which is why its percentage anomaly is large there. **b**, Trend in F_{ov}^S split into transport and salinity contributions with 95% block-bootstrap intervals, and the covariance residual, over each product’s own record and over the Argo era. **c**, Transport share of the change over every non-overlapping epoch pair the record supports (2901 for ORAS5, 151 for GLORYS12V1) at epoch lengths of 10, 13 and 15 years, treated as a sensitivity analysis over window choice. Ticks mark the median, 90th and 95th percentiles and the maximum; the percentage is the fraction of pairs below the dashed 0.25 line.

limb while the southward limb does not change. How much of that is a water-mass change rather than a redistribution of transport in depth differs between products: in GLORYS12V1 it is 94%, in ORAS5 only about two thirds, with the salinity-only part there not individually significant. The coarse upper-to-abyssal redistribution is small in both, and excluding the abyssal cell makes the salinification larger rather than smaller, so what remains in ORAS5 is a rearrangement of transport within the upper limb.

The part of this that observations can check, they support. Holding the circulation fixed and letting only the EN4 objective analysis vary, the northward limb salinifies at +0.026 to +0.028 PSU per decade ($p \leq 0.0005$) while the southward limb is flat. The limb asymmetry is therefore present in a salinity field that no model integrates forward, and the same construction returns a negative F_{ov}^S , -0.080 to -0.128 Sv, from observed salinity. The rate is less secure: ORAS5 runs about twice the observation-only value over the same years while GLORYS12V1 is within 30% of it, and part of that gap is expected because EN4 relaxes to climatology where profiles are sparse. The direction should be trusted; the ORAS5 rate should not. It also reframes what the diagnostic is sensing. A salinifying upper limb widens the limb contrast and drives F_{ov}^S downward regardless of what the overturning does, which is why F_{ov}^S and the transport are so weakly coupled on the timescales the records resolve. It is consistent in direction with the South Atlantic salinity accumulation attributed to a weakening overturning^{15,16}, but our decomposition locates the change in the transport-weighted upper limb rather than in the surface field, and we find no transport change to accompany it.

Two things temper this. The first is that F_{ov}^S is not the freshwater budget of the basin. The azonal

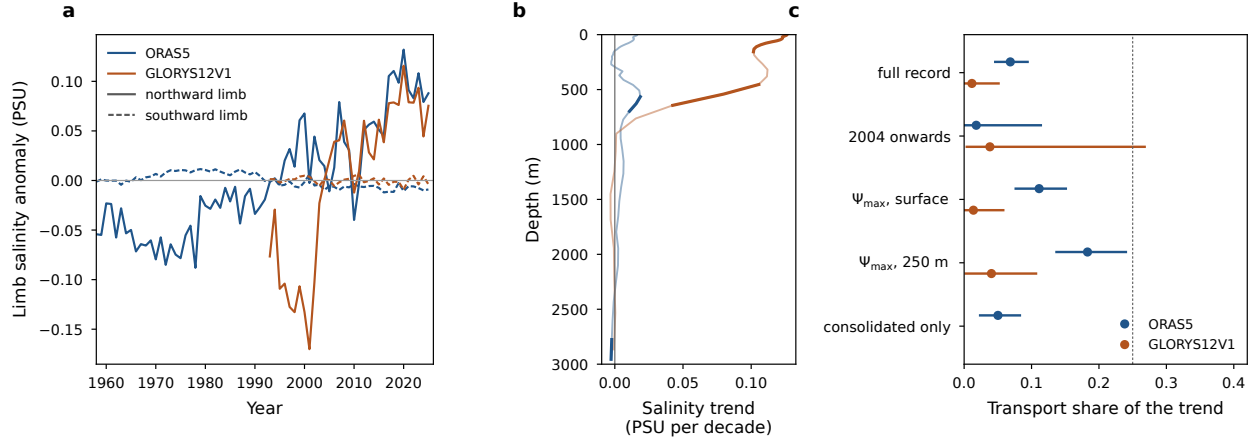


Figure 3. The northward limb salinifies; the southward limb does not. **a**, Transport-weighted salinity of the northward limb (solid) and the southward limb (dashed), as anomalies from their record means. **b**, Zonal-mean salinity trend against depth at the section over 1993–2025, drawn bold where the pointwise Santer-adjusted p is below 0.05. The panel is truncated at 3000 m; the abyssal limb discussed in the text begins below 4000 m. Adjacent levels are strongly correlated, so no field-significance claim is made from this panel. **c**, Transport share of the F_{ov}^S trend under every robustness test, with 95% block-bootstrap intervals: the full record, the Argo era, two alternative definitions of the overturning strength, and, for ORAS5, the homogeneous consolidated stream alone.

component at this section is several times larger, of the opposite sign, and trending upward by more than F_{ov}^S is trending downward: significantly so in ORAS5, and marginally in GLORYS12V1 ($p = 0.049$ on 8.5 effective degrees of freedom). Whether the basin as a whole is moving deeper into the self-reinforcing regime is therefore not settled by F_{ov}^S alone, and we make no claim that it is. Nor do we evaluate the northern boundary: the indicator in the theory we cite is the freshwater convergence between the southern and northern boundaries, and F_{ov}^S is a proxy for it whose adequacy we have not tested. Both are natural next steps and neither is addressed here.

The second is ECCO-V4r4, and it is a genuine disagreement rather than a caveat. It is the only product here that is dynamically consistent by construction and therefore the only one whose freshwater budget closes without assimilation increments, and it shows no decline in F_{ov}^S . Its limb salinities are flat only in the aggregate: its upper cell salinifies at +0.022 PSU per decade ($p = 0.025$), the same sign as the other two products, while a significant redistribution towards its abyssal cell cancels the effect. Against GLORYS12V1 this is not an underpowered null: on the window the two share, ECCO’s minimum detectable limb-salinity trend of 0.020 PSU per decade is well below the GLORYS12V1 rate of +0.083, so ECCO could have seen that salinification and did not. Against ORAS5, whose common-window rate is +0.019, the comparison is inconclusive. ECCO is also the one product in which the coarse limb redistribution is significant and acts to cancel a positive water-mass term, so its flat limb salinity is a cancellation rather than an absence of change. The mechanism we report is therefore a property of the two sequential-assimilation products, and it remains possible that what they are tracking is the arrival of the Argo array rather than a change in the ocean. The evidence against that reading is that the attribution is unchanged in every window and under every inho-

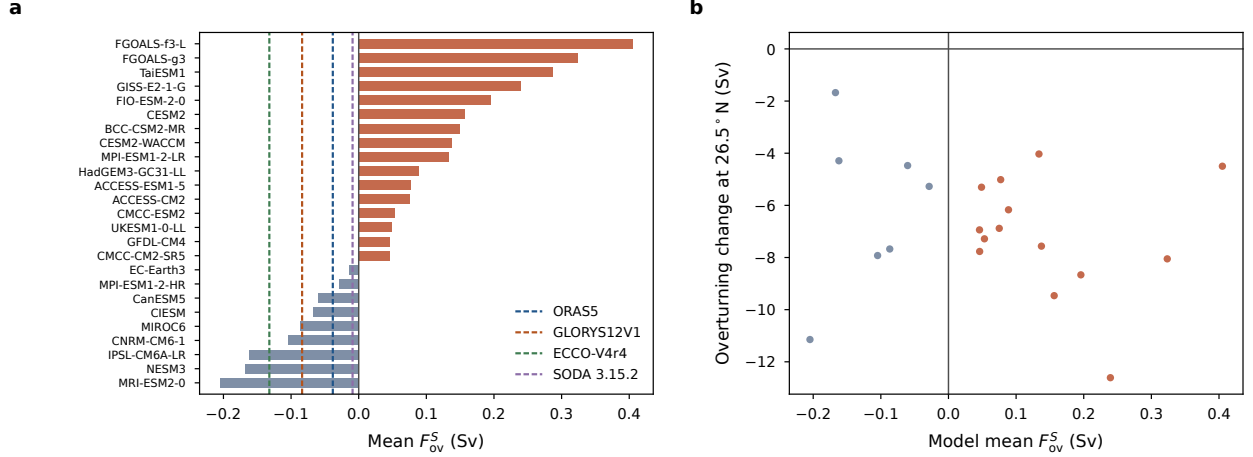


Figure 4. CMIP6 lies above the observational estimates, but cannot be sorted by the sign of F_{ov}^S . **a**, Mean F_{ov}^S at 34.5°S over 1950–1980 in 25 CMIP6 models, with the reanalysis estimates marked; SODA is shown for continuity with Fig. 1 and is not used quantitatively. No offset or bias correction is applied to either models or reanalyses. **b**, Model mean F_{ov}^S against the overturning change at 26.5°N from 1850–1900 to 2081–2100 in the 21 models with historical plus ssp585 projections ($r = -0.20$, $p = 0.40$, smallest reachable $|r| = 0.43$). This replaces a 19-year single-member trend test, whose null result is guaranteed by internal variability: 19-year trends span -3.5 to $+2.4$ Sv per decade across 50 members of the MPI-ESM1-2-LR large ensemble.

mogeneity correction we tried, and that EN4 reproduces the limb asymmetry independently. The evidence for it is that both sequential-assimilation products step at an observing-system boundary: GLORYS12V1’s decline is a level shift at the 2004 Argo transition rather than a trend, and the ORAS5 record steps across its consolidated-to-operational join, significantly so in the limb salinity itself ($+0.038$ PSU, $p = 0.029$), after which its Argo-era trend is no longer significant. Neither product’s *rate* of decline should be taken at face value. We could not obtain the salinity assimilation increments that would settle the question directly, and we regard it as open.

Beyond that, all of these products are forced by the ERA family, so cross-product agreement is a weaker form of independence than the product count suggests: they differ in ocean model and in assimilation scheme, but not in the atmosphere driving them. The Copernicus distribution of ORAS5 publishes one of the system’s five ensemble members¹⁹, so cross-product spread has to stand in for within-product spread throughout, and that spread, 0.094 Sv, exceeds even the widest within-product interval²⁰. We compute ECCO on the distributed 0.5° interpolation rather than the native LLC90 grid, which does not conserve transport exactly, and since ECCO is now the load-bearing counterexample that limitation matters more than it would if it were one supporting vote among four. The three products differ by 46% in northward-limb transport at this section (14.6 to 21.3 Sv), and we have not applied a published SAMBA or SAMOC mean to arbitrate between them. That omission is not neutral: the weakest of the three is ECCO, which is also our counterexample, so a comparison against the observed mean transport could as easily undermine the counterexample as support it, and it should be done before these numbers are used to weigh one product against another. The reanalyses are also least constrained by observations exactly where we evaluate them,

since the observing network at 34.5°S is far sparser than the sustained array at 26.5°N^{21–24}.

For models the implication cuts two ways. Correcting the freshwater transport bias is necessary if a model is to represent the salt-advection feedback with the right sign, and the CMIP6 median sits above every observational estimate we computed. But the majority count usually quoted is not distinguishable from a coin flip once model genealogy is accounted for, and we find no detectable relationship between the sign of F_{ov}^S and the rate at which a model’s overturning weakens, either over the satellite era or over the forced response to 2100, with a sample that could not have detected a moderate one. What we do find is that model F_{ov}^S is strongly correlated with pre-industrial overturning strength, which suggests that the freshwater transport bias is partly a symptom of the overturning strength bias. If that is right, correcting F_{ov}^S in isolation is the wrong target, and the two biases should be addressed together.

What would settle the transient question is observational rather than analytical. The claim we make is now a claim about limb salinities, which sustained hydrography at 34.5°S and the repeat sections that cross it can test directly, without passing through an assimilating model; projecting an observation-only salinity analysis onto the same limb mask is the single most informative next step and we regard it as the test this result should be judged by. Sustained transport monitoring at this latitude, extended towards the length RAPID has reached at 26.5°N, would constrain both factors of equation (1) at once. Reanalyses forced outside the ERA family would test how much of the present agreement comes from shared boundary conditions, and release of all five ORAS5 ensemble members would separate assimilation noise from ocean signal within a single product. Until then the defensible statement is the narrow one: the Atlantic freshwater budget sits where the salt-advection feedback is self-reinforcing, and in the products that decline, the recent decline in that diagnostic records a salinifying upper limb rather than a slowing circulation.

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Methods

Ocean reanalyses

Three reanalyses are computed here from monthly fields (Extended Data Table 1). ORAS5¹⁹ is a NEMO configuration at 0.25° with 75 vertical levels, assimilated by NEMOVAR 3D-Var FGAT and forced by ERA-40, ERA-Interim and ERA5; we use 816 gap-free monthly fields of meridional velocity and salinity covering 1958–2025. The Copernicus distribution joins the consolidated reanalysis, which ends in 2014, to the operational stream thereafter, and the two are treated separately wherever it matters. GLORYS12V1²⁵ is a NEMO configuration at 1/12° with 50 levels, assimilated by a reduced-order Kalman filter and forced by ERA-Interim and ERA5; we use 396 monthly fields covering 1993–2025. ECCO-V4r4²⁶ is an MITgcm state estimate on the LLC90 grid, nominally 1°, fitted to observations by the adjoint method and therefore dynamically consistent by construction, with no assimilation increments; its atmospheric forcing is ERA-Interim plus estimated control adjustments. We analyse the distributed 0.5° lat-lon interpolation with 50 levels, using all 312 monthly fields covering 1992–2017. Interpolation from the native grid does not conserve volume transport exactly, which is a limitation we state rather than correct.

SODA 3.15.2²⁷ is shown in Fig. 1a for context but is not used in any quantitative claim. It is a MOM5 and SIS configuration run at 1/4°, assimilated by optimal interpolation with bias correction and forced by ERA5, and the archive we can obtain provides annual rather than monthly fields on a regridded 0.5° Mercator grid. Because F_{ov}^S is bilinear in velocity and salinity, forming it from annual means omits the within-year covariance; measured in ECCO, where both routes are available, the omission shifts F_{ov}^S by +0.0124 Sv, exceeding SODA’s record mean of −0.0091 Sv in magnitude. The corresponding shifts are +0.0000 Sv in ORAS5 and −0.0013 Sv in GLORYS12V1.

The section

At the grid row nearest 34.5°S,

$$F_{ov}^S = -\frac{1}{S_0} \int_{-H}^0 V_{bc}(z) [\bar{S}(z) - S_0] dz, \quad (2)$$

where $V_{bc}(z)$ is the zonally integrated meridional velocity after removal of the net section transport, $\bar{S}(z)$ the width-weighted zonally averaged salinity and $S_0 = 35$ PSU. A negative F_{ov}^S means the overturning imports freshwater, equivalently that it exports salt from the Atlantic.

The Atlantic segment at each row is the contiguous run of ocean points containing 25°W, rather than a fixed longitude box. This matters. A box from 70°W to 20°E at 34.5°S admits, in GLORYS12V1, a second water body of about 20 grid points near 58 to 57°W whose surface salinity is 0.00 to 0.04 PSU. Those values are too fresh for the outer Río de la Plata, which is brackish rather than fresh, so they are most likely estuary, lake or land-mask cells rather than a resolved plume; either way they are not part of the open Atlantic along this row. Including them depresses $\bar{S}$ in the upper cells and shifts F_{ov}^S by +0.013 Sv, about a sixth of the GLORYS12V1 record mean. At the 0.5° resolution of ECCO the

feature is unresolved and the contiguity test makes no difference. The resulting sections contain 294 ocean points at 34.56°S in ORAS5, 881 at 34.50°S in GLORYS12V1 and 146 at 34.75°S in ECCO; these are counts of ocean points, not of the width of a longitude box.

The net transport removed is -1.22 Sv in ORAS5, -1.17 in GLORYS12V1 and -0.98 in ECCO, and removing it shifts F_{ov}^S by $+0.0086$, $+0.0083$ and $+0.0066$ Sv. It is of the sign and order expected for the Bering Strait inflow plus net precipitation minus evaporation over the basin to the north, but we do not claim it is that quantity: across the latitude ladder it varies non-monotonically by 1.1 Sv in ORAS5 and 2.5 Sv in GLORYS12V1, far more than the precipitation minus evaporation of the intervening bands, so a substantial part of it is section construction, grid metrics or genuine mass imbalance. The justification for removing it is that an overturning component of the freshwater transport is not defined without doing so, and that equation (2) is otherwise sensitive to S_0 .

The azonal component is $F_{az} = -(1/S_0) \iint v' S' dx dz$, with primes the departures from the width-weighted zonal mean at each depth. It is computed from monthly means and therefore excludes sub-monthly covariance.

The limb factorisation

Because $\int V_{bc} dz = 0$, the depth levels where $V_{bc} > 0$ carry a volume transport T exactly balanced by $-T$ in the levels where $V_{bc} < 0$. Defining the transport-weighted limb salinities

$$S_{\uparrow} = \frac{1}{T} \int_{V_{bc} > 0} V_{bc} \bar{S} dz, \quad S_{\downarrow} = \frac{1}{T} \int_{V_{bc} < 0} (-V_{bc}) \bar{S} dz, \quad (3)$$

equation (2) becomes $F_{ov}^S = -T(S_{\uparrow} - S_{\downarrow})/S_0$ exactly, because the $-S_0$ term integrates to zero. Both factors are computed directly from the section profile and neither is defined in terms of the other, which is what distinguishes this from writing $F_{ov}^S = -\Psi \Delta_v S / S_0$ with $\Delta_v S \equiv -S_0 F_{ov}^S / \Psi$. In that construction the second factor is the first divided out; it absorbs every change in the vertical structure of the velocity as well as every change in salinity, it closes to machine precision by construction, and the implied overturning share is fixed by the algebraic identity $(\Psi \text{ term}) / \Delta F_{ov}^S = (F_{\text{early}} / \Delta F_{ov}^S) (\delta \Psi / \Psi_{\text{early}})$, where F_{early} and Ψ_{early} are the base-epoch values. We do not use it except as a labelled sensitivity test, and we note that because that construction pins the share to the ratio of fractional changes it cannot return a large value whatever the physics. The residual of the limb factorisation is below 1.5×10^{-14} Sv, which is double-precision round-off and confirms only that the net transport removal closed.

Attribution

For a change between two epochs we use the symmetric decomposition

$$\Delta F_{ov}^S = -\frac{1}{S_0} \left(\overline{\Delta S} \Delta T + \bar{T} \Delta(\Delta S) \right), \quad (4)$$

with $\bar{T}$ and $\overline{\Delta S}$ the means of the two epochs. This is exact, has no cross term to assign, and does not depend on which epoch is taken as the base point. The asymmetric early-base and late-base

expansions differ from it by plus and minus half the cross term and are reported alongside.

The primary attribution is time-continuous. Holding one factor at its record mean gives two counterfactual series, $F_T(t) = -T(t)\overline{\Delta S}/S_0$ and $F_{\Delta S}(t) = -\overline{T}\Delta S(t)/S_0$. Unlike equation (4) this split is *not* exact: the trends of the two counterfactuals differ from the F_{ov}^S trend by the trend of the transport-salinity covariance, and that residual is reported for every case in the Results and in Fig. 2b. It is 3% of the transport term in ORAS5 over its own record and 108% in GLORYS12V1, so for GLORYS12V1 the transport contribution is quoted as not distinguishable from zero rather than as a point value. The transport share is $|\text{trend}(F_T)|/|\text{trend}(F_{ov}^S)|$, quoted only where the total trend is resolvable, because a ratio with an unresolved denominator is not identified. The numerator and denominator are bootstrapped jointly, using the same block indices, so the ratio interval is coherent. Two cases are excluded by that rule and reported here for completeness: the ORAS5 common-window share is 0.368 [0.093, 6.19], and the ECCO census yields one usable pair out of 29.

The epoch census evaluates equation (4) for every non-overlapping early and late epoch pair a record supports at epoch lengths of 10, 13 and 15 years, discarding pairs whose total change is below 5 mSv in magnitude, and reports the distribution of shares. These pairs are overlapping views of one series, not independent samples, and are used as a sensitivity analysis over window choice.

The overturning strength is also computed as the maximum of the depth-space overturning streamfunction, from the surface and with the integration restarted at 250 m, and the attribution repeated with each. Those streamfunctions are cumulated from the uncorrected velocity, so that definition is independent of how the net section transport is removed.

Limb composition

The sign of V_{bc} changes more than once, so the northward limb comprises an upper cell and, beneath the southward deep water, a northward abyssal cell. Writing $S_{\uparrow} = \sum_i w_i S_i$ over these sub-limbs with weights $w_i = T_i/T$ summing to one, holding the weights at their record means isolates the water-mass change and holding the sub-limb salinities at theirs isolates the redistribution, exactly as in the main attribution. The decomposition is evaluated on monthly fields, so the sub-limb transports sum to the same T used elsewhere. The reconstructed $S_{\uparrow}$ trend, +0.027 PSU per decade in ORAS5, still differs by about 7% from the directly computed +0.025, because a mean of monthly products is not the product of monthly means. Identifying the sub-limbs on annual-mean profiles instead smooths out the deep sign reversals, understates the abyssal cell by 26 to 30% and leaves a shortfall against T of 5.4%, 4.1% and 1.5%; that route is not used. Vertical cell thicknesses are taken from each product's own grid bounds rather than from differences of the cell-centre depths, which for ECCO differ by 228 m in the column total.

Independently, the section is truncated at the deep zero crossing so that the abyssal cell is excluded, the net-transport removal is re-applied within the truncated layer so that the factorisation is again exact, and F_{ov}^S , T and ΔS are recomputed for the upper cell alone.

Frozen-field and observation-only tests

ΔS is transport-weighted, so it responds both to the salinity field and to a redistribution of transport in depth. The two are separated by recomputing the limb salinities with $\bar{S}(z)$ frozen at its record mean while $V_{bc}(z, t)$ varies, and again with $V_{bc}(z)$ frozen while $\bar{S}(z, t)$ varies. This is done at the full vertical resolution of the section, so it captures redistribution within a limb as well as between limbs. The sub-limb decomposition described above resolves only the coarse upper-to-abyssal part and is reported alongside it; it is evaluated on monthly fields so that the sub-limb transports sum to the same T used elsewhere.

For the observation-only test the velocity structure is frozen at each reanalysis record-mean profile, interpolated onto the EN4.2.2 levels and renormalised so that its depth integral again vanishes, and the limb salinities are formed from the EN4 objective analysis of profile observations (g10 corrections) over 2005–2024, on a contiguous section built the same way. EN4 is an optimal interpolation of profiles onto a 1° grid with no forward model, so a trend computed this way is a trend in observed water masses under a fixed circulation. Because EN4 is on whole-degree latitudes we use the 34°S row: 34.5°S rounds to 35°S , which lies south of Cape Agulhas (34.83°S), where Africa no longer closes the basin and a contiguous section runs east across the Indian Ocean to Australia. EN4 relaxes towards climatology where profiles are sparse and core Argo does not sample below 2000 m, so its trends are lower bounds.

Sensitivity to the removal of the net transport

Because the limb split depends on where V_{bc} changes sign, and that depends on how the net section transport is removed, the whole analysis is repeated with the net transport removed in proportion to $|V|$ rather than uniformly over the cross-sectional area. T changes by less than 0.6% and the transport share by less than 0.01 in both eddy-permitting products.

Statistics

Trends are ordinary least squares on annual means. Significance uses the effective sample size of Santer et al.²⁸: with r_1 the lag-1 autocorrelation of the regression residuals, $N_{\text{eff}} = N(1 - r_1)/(1 + r_1)$, the slope standard error is inflated by $\sqrt{(N - 2)/(N_{\text{eff}} - 2)}$ and the t -test uses $N_{\text{eff}} - 2$ degrees of freedom. N_{eff} is reported alongside adjusted p -values in the Results, because several load-bearing tests have few effective degrees of freedom; the GLORYS12V1 limb-salinity trend has $N_{\text{eff}} = 6.9$. For every window in which we report a null result we also give the minimum detectable trend, the smallest slope that window could have resolved at the 5% level, so that an absence of significance is not read as an absence of change. Correlation detection limits are quoted as the smallest $|r|$ reaching $p < 0.05$, that is at 50% power; the 80% power thresholds are larger.

Trend confidence intervals come from a circular moving-block bootstrap of the regression residuals with 10,000 iterations, added back to the fitted line and refitted; resampling the values rather than the residuals would destroy the trend and return a null distribution. Intervals on record means use

a circular moving-block bootstrap of the values, reported for block lengths of 2, 3, 5, 10, 15 and 20 years, with the 5-year value quoted in the text. Since a mean over a trending record is not an estimate of a stationary population quantity, the mean over the most recent decade is reported alongside it.

Where a record contains a known inhomogeneity, trends are refitted as $F(t) = a + bt + cI(t \geq t_{\text{splice}})$, with I the indicator function, so that the level shift and the trend are estimated jointly, with standard errors inflated by the same Santer factor computed from the residuals of that fit. For ORAS5, $t_{\text{splice}} = 2015$. Estimating the step by differencing stream means instead conflates it with the trend and gives a larger value, which is why the regression estimate is the one quoted.

Per-depth salinity trends in Fig. 3b are marked at the pointwise 5% level; adjacent levels are strongly correlated, so no field-significance claim is made from them and none is used in the argument.

CMIP6 and observed transports

Mean F_{ov}^S at 34.5°S is computed for 25 CMIP6²⁹ historical simulations over 1950–1980 on their native grids by equation (2), one realisation per model. No offset or bias correction of any kind is applied, to either the models or the reanalyses. Model genealogy is accounted for by grouping close relatives (CESM2 with CESM2-WACCM and TaiESM1, the two MPI-ESM1-2 resolutions, the two ACCESS models, the two FGOALS models, the two CMCC models, and UKESM1-0-LL with HadGEM3-GC31-LL) and retaining the member whose F_{ov}^S is median within each family. Only the MPI family is internally inconsistent in sign, and the count is reported for both of its members. Sign counts are tested against a binomial null. Two provenances of the model F_{ov}^S were in circulation in earlier work and differ by up to 0.041 Sv (root mean square 0.016 Sv over 16 models, worst case MPI-ESM1-2-LR) without changing any sign; everything here uses the decomposition summary alone.

Overturning at 26.5°N is taken from `msftmz`, or `msfty` where `msftmz` is unavailable, on the `atlantic_arctic_ocean` basin of the native `gn` grid for variant `r111p1f1`, as the maximum over depths at or below 500 m of annual means, with historical and `ssp585` concatenated. Trends over 2005–2023 are available for 16 models and the change from 1850–1900 to 2081–2100 for 21. The internal-variability floor for a 19-year trend is taken from 50 members of the MPI-ESM1-2-LR large ensemble. Leave-one-out ranges are reported for all three correlation tests.

The RAPID array²² gives -0.889 Sv per decade at 26.5°N over 2005–2023, with an unadjusted p of 0.093 that becomes 0.205 after the effective-sample-size correction ($N_{\text{eff}} = 11.9$). We do not compare the modelled mean transports against the SAMBA array at 34.5°S^{21,24}. The relevant published quantity would be a mean rather than a trend, and its absence here is a limitation rather than a consequence of the array’s length.

Data availability

ORAS5 is distributed by the Copernicus Climate Data Store, GLORYS12V1 by the Copernicus Marine Service, ECCO-V4r4 by NASA PO.DAAC, SODA 3.15.2 by the University of Maryland, RAPID by the RAPID-AMOC programme and CMIP6 output through the Earth System Grid

689 Federation. Dataset identifiers, version strings and access dates are listed in the archived code
690 repository.

691 **Code availability**

692 Analysis and figure code is archived with the manuscript. The section profiles and the limb factorisa-
693 tion are computed by `compute_paper3v2_section_profiles.py`, the attribution and its robustness
694 tests by `analysis_paper3v2_attribution.py`, the limb composition by `analysis_paper3v2_limb_composition.py`,
695 the splice and fractional-rate analysis by `analysis_paper3v2_stepfit.py`, the latitude sensitivity by
696 `analysis_paper3v2_latitude_sensitivity.py`, the CMIP6 analysis by `analysis_paper3v2_cmip6.py`,
697 and each figure by the correspondingly named `plot_paper3v2_*.py` script.

Extended Data

Extended Data Fig. 1 | Surface salinity trends, 1993–2025. Per-pixel trends on deseasonalised monthly anomalies for ORAS5 and GLORYS12V1, hatched where the Santer-adjusted p exceeds 0.05. The effective-sample-size correction reduces the significant ocean area from 76% to 36% in ORAS5 and from 76% to 51% in GLORYS12V1. Atlantic band means disagree in sign between the products in all three bands examined (60°N to 80°N, 40°N to 60°N, 10°S to 35°S), and only one of the six band-and-product combinations is significant, GLORYS12V1 over 10°S to 35°S (+0.061 PSU per decade, $p = 0.0002$). There is no significant subpolar North Atlantic freshening in either product. Surface salinity trend patterns are in any case ambiguous, since amplification of the water cycle and a uniform surface heat flux both produce patterns resembling the one that would be attributed to ocean transport³⁰, which is why the argument in the main text rests on the transport-weighted limb salinities at the section and these maps are not used as evidence.

Extended Data Fig. 2 | Section profiles and the structure of the limbs. **a**, Time-mean zonally integrated meridional velocity after removal of the net section transport, over the matched 1993–2017 window. The sign changes of this profile define the limbs. **b**, Time-mean zonally averaged salinity over the same window. **c**, The northward-limb transport split into its upper cell (solid) and its abyssal cell (dashed), on monthly fields so that the two sum to T . The barotropic-corrected velocity changes sign three to seven times in ORAS5 (modally four), three to four in GLORYS12V1 and three to five in ECCO; the upper limb reaches to about 1325, 1245 and 1318 m respectively, and the abyssal limb begins at about 4044, 4405 and 4264 m. The abyssal cell carries 3.8, 4.8 and 2.2 Sv, that is 20%, 23% and 15% of the northward limb, and is 0.21 to 0.49 PSU fresher than the upper cell, which is why the redistribution between them has to be excluded before the widening contrast can be read as a water-mass change. Its transport trend is significant in ECCO (+0.36 Sv per decade, $p = 0.041$) and in ORAS5 (+0.17, $p = 0.0004$), and not in GLORYS12V1.

Extended Data Table 1 | Reanalysis products.

	ORAS5	GLORYS12V1	ECCO-V4r4	SODA 3.15.2
Ocean model	NEMO	NEMO	MITgcm	MOM5, SIS
Model resolution	0.25°	1/12°	LLC90, $\sim 1^\circ$	0.25°
Grid analysed	native	native	0.5° lat-lon	0.5° Mercator
Vertical levels	75	50	50	50
Assimilation	NEMOVAR 3D-Var	reduced-order	adjoint	optimal
	FGAT	Kalman filter	(no increments)	interpolation
Atmospheric forcing	ERA-40, ERA-Interim, ERA5	ERA-Interim, ERA5	ERA-Interim (adjusted)	ERA5 (COARE4)
Fields used	816 monthly	396 monthly	312 monthly	43 annual
T , 1993–2017 (Sv)	17.9	21.1	14.6	not recomputed
Record	1958–2025	1993–2025	1992–2017	1980–2022
Ocean points	294	881	146	not recomputed
T (Sv)	18.7	21.3	14.6	not recomputed
ΔS (PSU)	+0.068	+0.136	+0.307	not recomputed
Used quantitatively	yes	yes	yes	no

